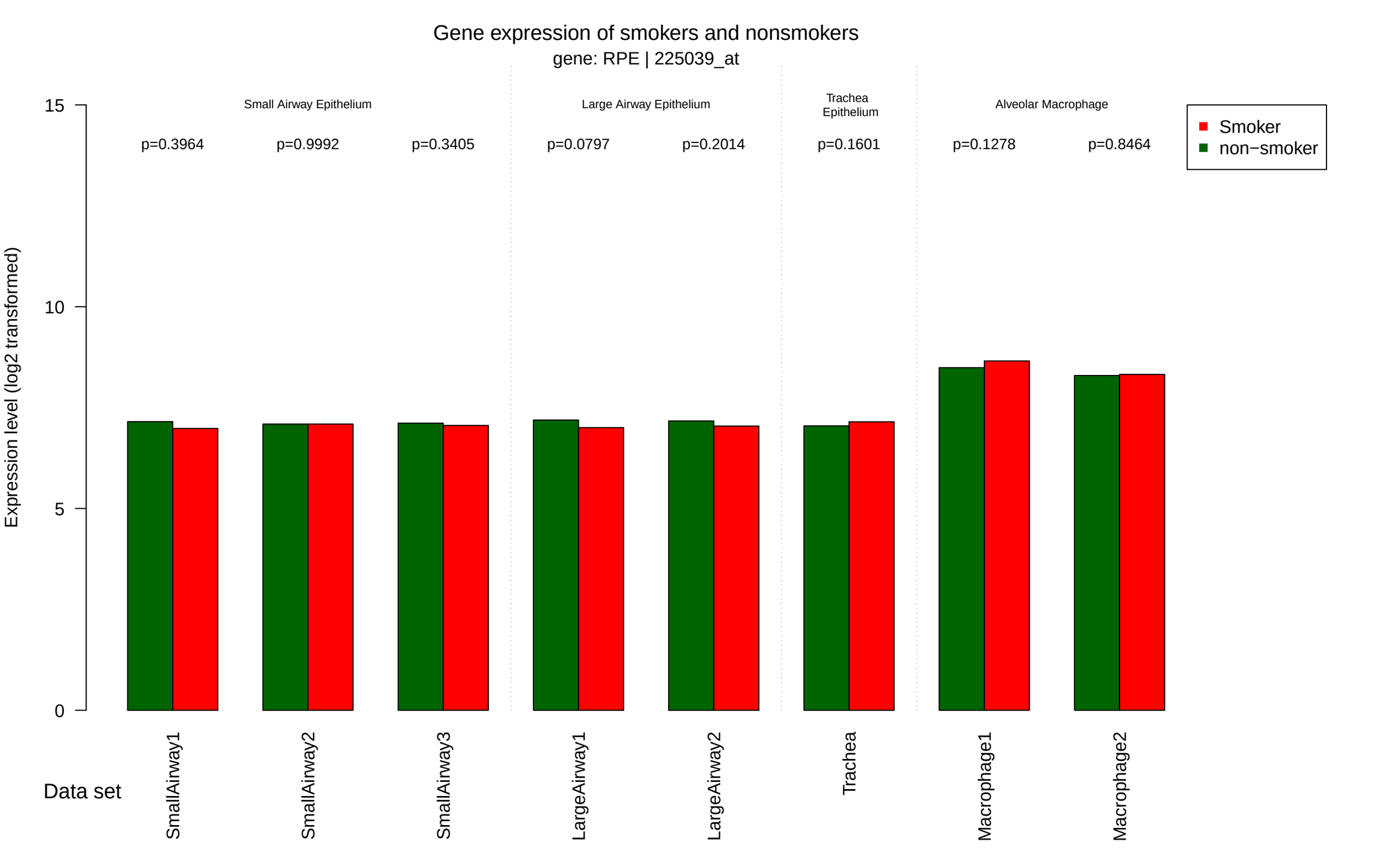
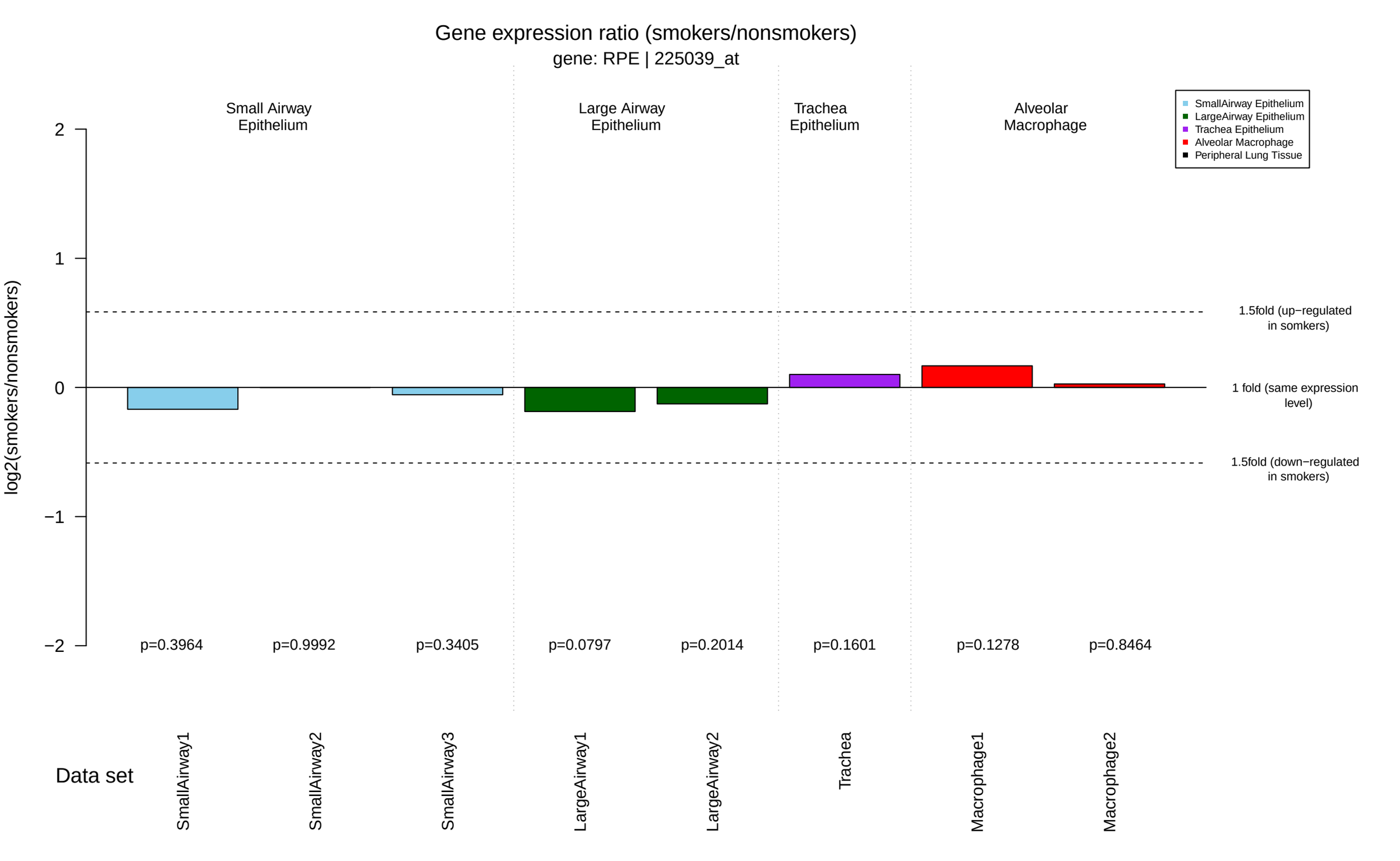
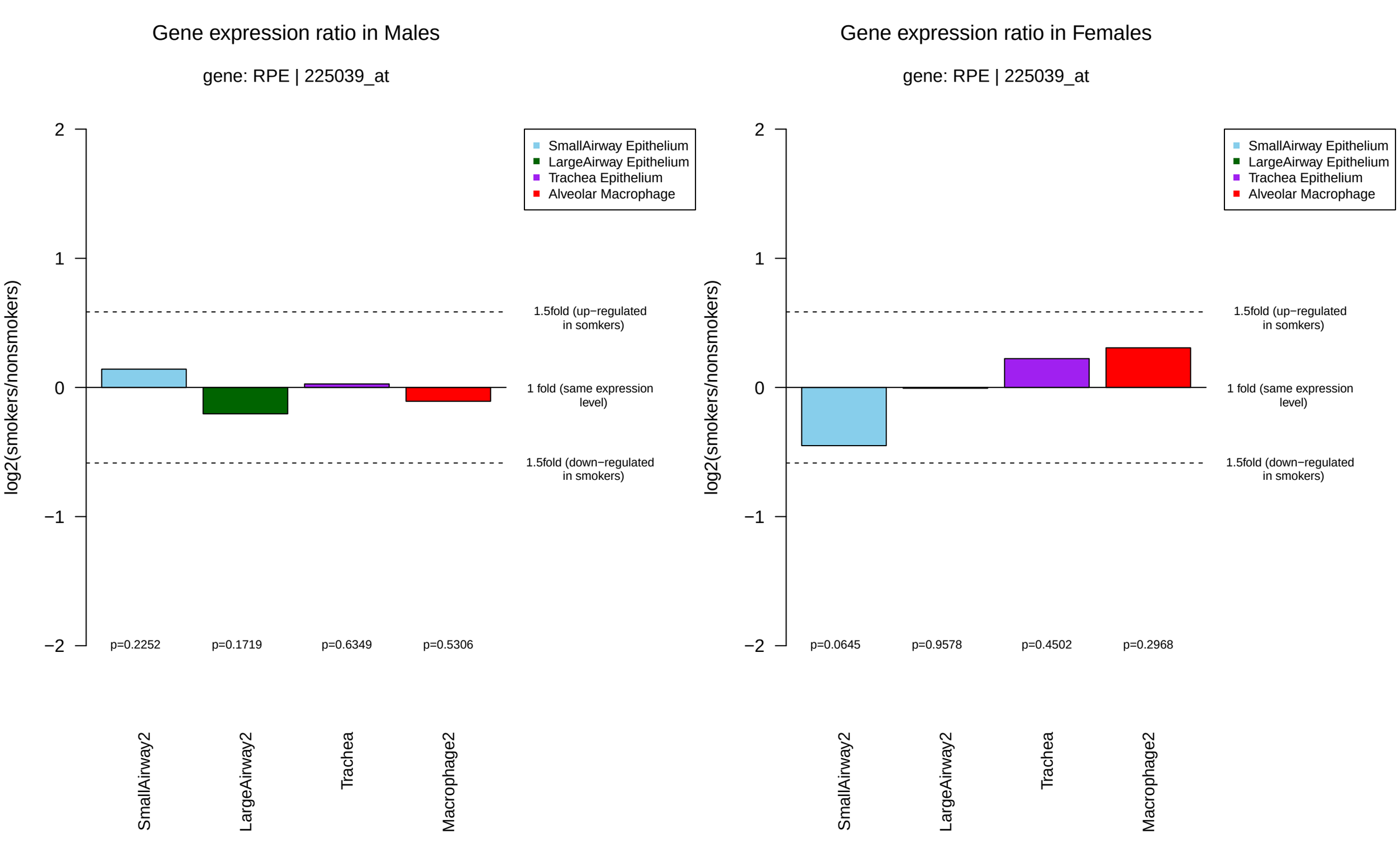
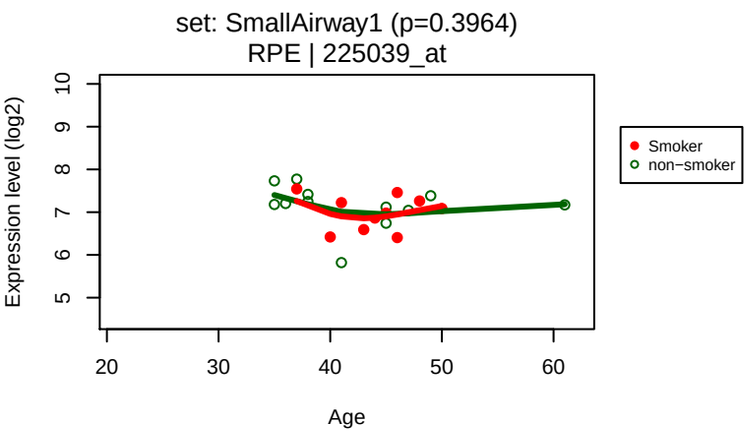
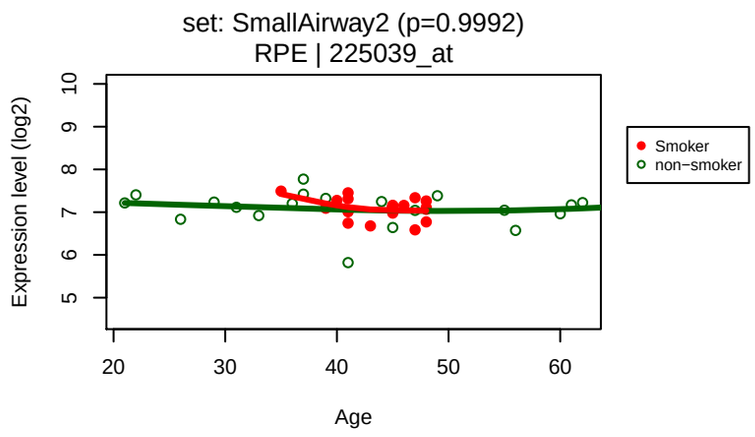
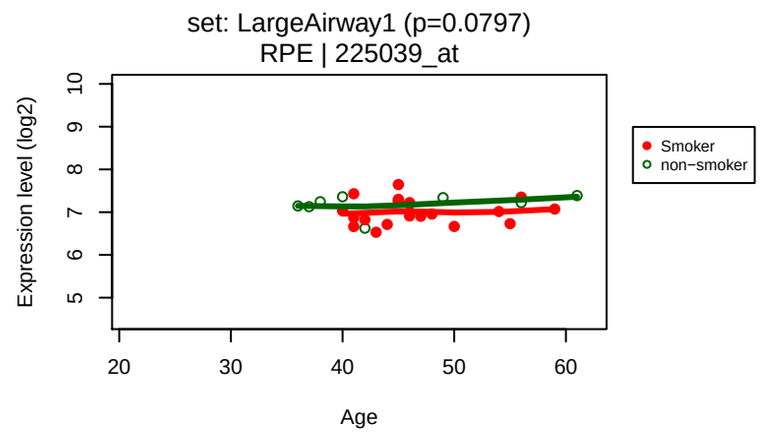

Gene expression of smokers and nonsmokers
gene: RPE | 225039_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3964
p=0.9992
p=0.3405
p=0.0797
p=0.2014
p=0.1601
p=0.1278
p=0.8464
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

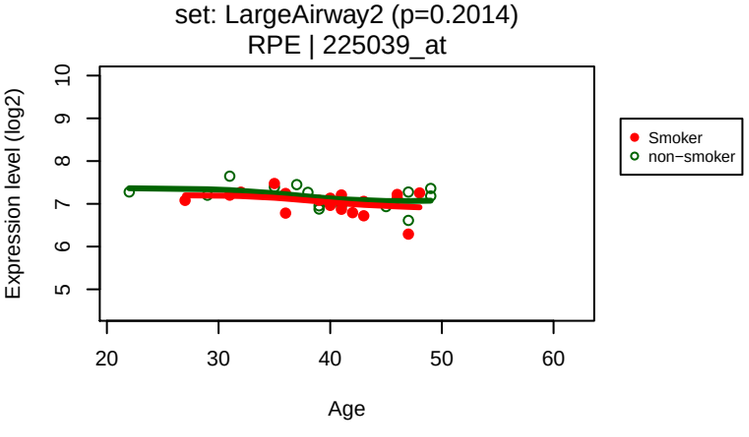

Gene expression ratio (smokers/nonsmokers)
gene: RPE | 225039_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3964
p=0.9992
p=0.3405
p=0.0797
p=0.2014
p=0.1601
p=0.1278
p=0.8464
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

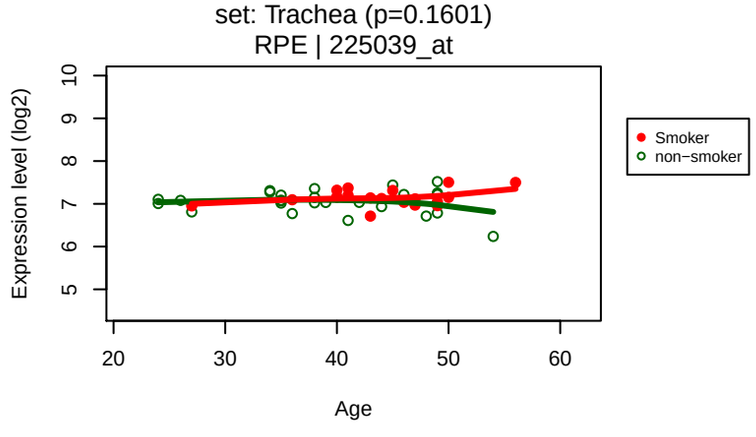

Gene expression ratio in Males
gene: RPE | 225039_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2252
p=0.1719
p=0.6349
p=0.5306
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPE | 225039_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0645
p=0.9578
p=0.4502
p=0.2968
SmallAirway2
LargeAirway2
Trachea
Macrophage2

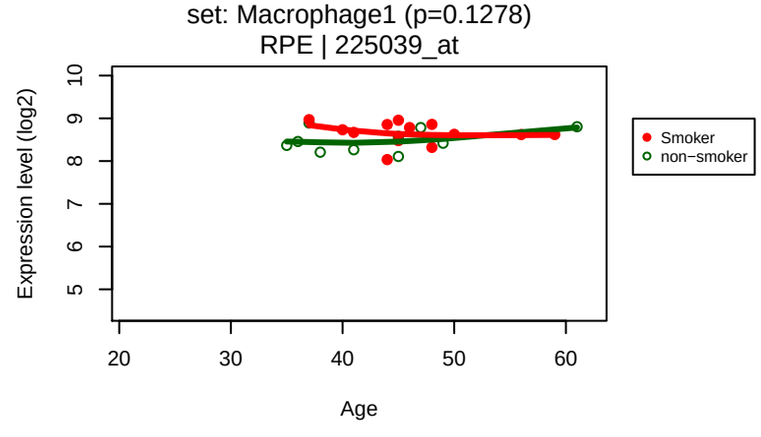

set: SmallAirway1 (p=0.3964)
RPE | 225039_at
Expression level (log2)
Age
Smoker
non−smoker

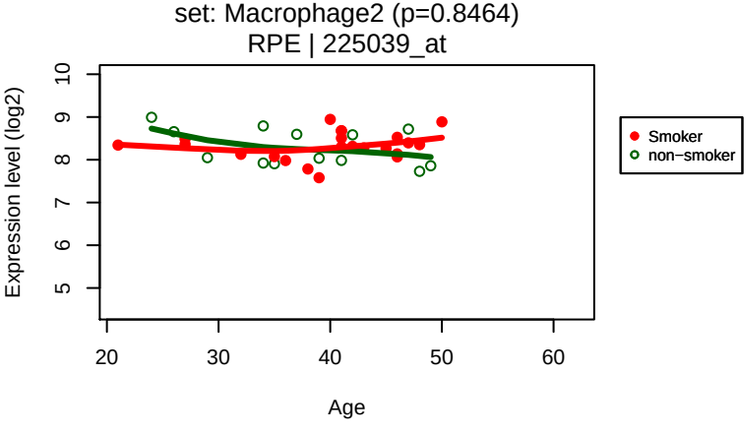

set: SmallAirway2 (p=0.9992)
RPE | 225039_at
Expression level (log2)
Age
Smoker
non−smoker

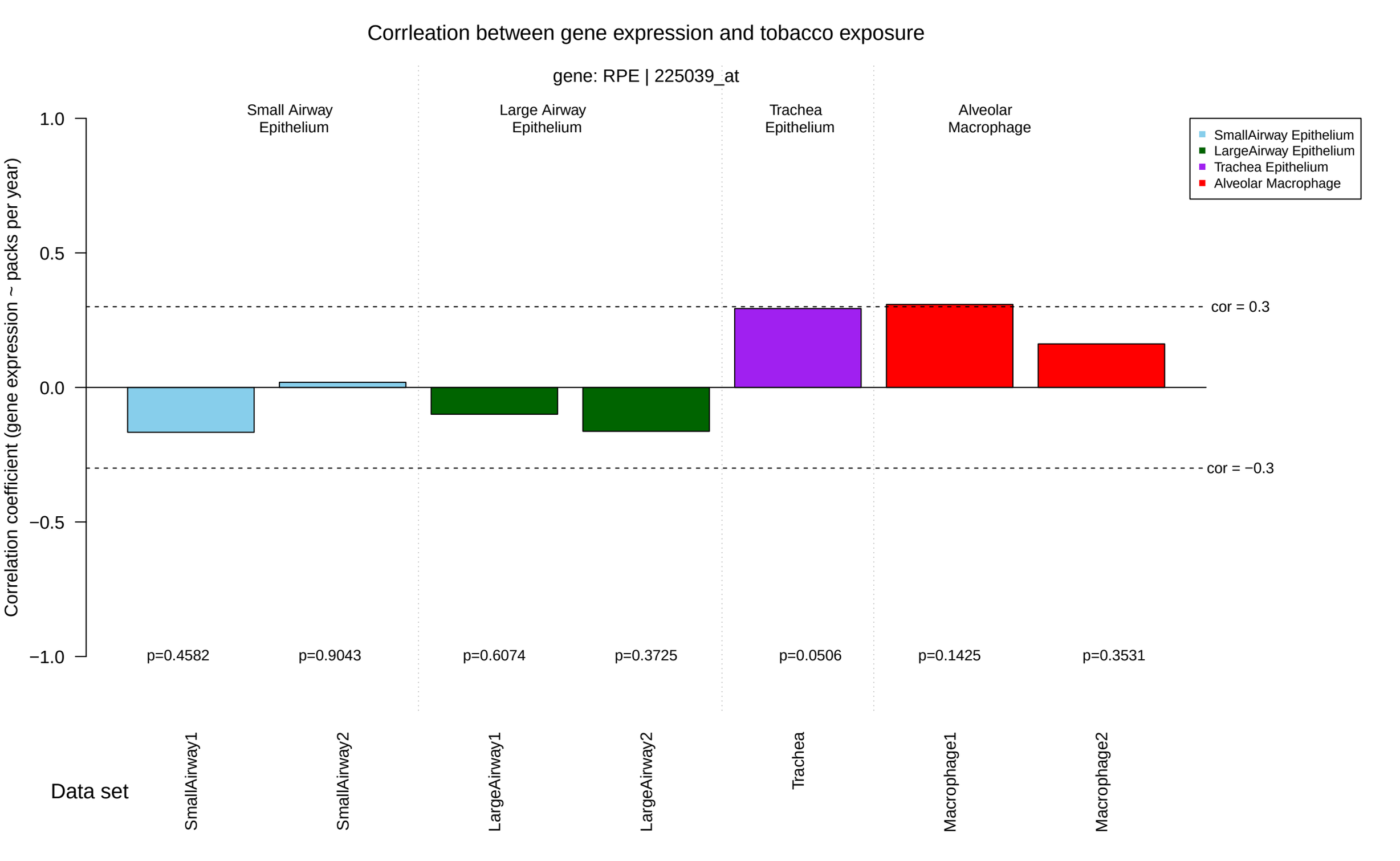

set: LargeAirway1 (p=0.0797)
RPE | 225039_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2014)
RPE | 225039_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1601)
RPE | 225039_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1278)
RPE | 225039_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8464)
RPE | 225039_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPE | 225039_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4582
p=0.9043
p=0.6074
p=0.3725
p=0.0506
p=0.1425
p=0.3531
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2